

Project Support Session

Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

Project Problem Statement

Business Context

A healthcare provider network offers a patient portal for accessing EHR data (labs, medications, visit summaries, imaging impressions, discharge instructions). Patients frequently misunderstand clinical terminology and next steps, driving high volumes of portal messages and call-center contacts, reduced adherence, and dissatisfaction. Patients also search online, encountering misinformation that may conflict with their records.

However, a patient-facing AI layer introduces non-trivial risks:

- misinterpretation of outputs as medical advice
- unsafe self-management behaviors
- inequities from varying health literacy, and
- PHI exposure through new interfaces and integrations.

The organization therefore needs a patient education and navigation assistant that is clinically safe-by-design, privacy-preserving, and auditable, with clear boundaries and escalation to clinicians.

Project Problem Statement

Objective

The objective is to build a POC in the form of a guardrailed, patient-facing AI education and navigation assistant powered by a single-agent system within the patient portal that:

- explains selected parts of the patient's record in plain language,
- answers general medical questions using vetted sources with citations, and
- enforces safety constraints (no diagnosis, no medication changes, no urgent-care triage beyond 'seek immediate care')

The end goal is to scale and expose this POC as an assistant to the patients to reduce avoidable support contacts and improve comprehension/adherence.

Project Problem Statement

Test Queries

1. *What does my Hemoglobin A1c result mean?*
2. *What is atorvastatin used for, and what are common side effects?*
3. *Can you summarize my most recent visit note and list the follow-up instructions?*
4. *My creatinine is high. Should I stop my lisinopril?*
5. *I'm having chest tightness today and I saw my lab report shows high potassium. Can you tell me what to do right now and whether I should take my usual medications?*

Dataset Overview

- A structured backend modeled as a simplified Electronic Health Record (EHR) system to support operational clinical data storage and retrieval.

Database table Name	Database table Content
Patients	Identity & personalization (minimal PHI)
Encounters	Visit context and care details
Clinical Notes	Free-text documentation
Labs	Results with reference ranges & interpretation flags
Medications	Active and historical prescriptions
Allergies	Reaction and severity tracking

Dataset Overview

- Curated structured files supporting education, citation control, and safety governance.

CSV File Name	File Content
Trusted sources catalog	catalog of approved sources and citation metadata.
Patient-friendly lab explanations	plain-language explanations for common lab tests.
Medication education content	medication education content
Safety policy rules	safety rules and escalation/refusal triggers.



Project Evaluation Rubric

Section	Points
Data Loading and Model Initialization	5
Defining Tools	18
Defining the Agent	15
Test Cases	10
Conclusions and Business Recommendations	4
Presentation/Notebook - Overall Quality	8

Project Evaluation Rubric

Data Loading and
Model Initialization

Defining Tools

Defining the Agent

Test Cases

Conclusions and
Business
Recommendations

Presentation/Notebook - Overall Quality

What to Cover?

Load the SQLite database

Load the CSV files

Guiding Questions and Considerations

How do we verify that the database has been loaded correctly (e.g., listing tables, running a simple SELECT query)?

How do we verify that each CSV has been loaded properly (e.g., checking shape, column names, a few sample rows)?

Project Evaluation Rubric

Data Loading and
Model Initialization

Defining Tools

Defining the Agent

Test Cases

Conclusions and
Business
Recommendations

Presentation/Notebook - Overall Quality

What to Cover?

Set up the LLM

Guiding Questions and Considerations

How do we choose the appropriate OpenAI model (e.g., gpt-4o, gpt-4o-mini) based on the task at hand and credits constraints?

How do we securely manage and load the OpenAI API key?

How do we configure key parameters (e.g., temperature, max_tokens, top_p) when calling the OpenAI model to suit the use case?

Project Evaluation Rubric

Data Loading and
Model Initialization

Defining Tools

Defining the Agent

Test Cases

Conclusions and
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Presentation/Notebook - Overall Quality

What to Cover?

Define a list of tools to be used
by the AI Agent

Guiding Questions and Considerations

How do we identify which tasks requires a deterministic vs a probabilistic (LLM-based) execution?

How do we choose a minimal yet sufficient set of tools that cover the required capabilities without making the agent overly complex?

How do we clearly define each tool's inputs, outputs, and purpose so the agent can select and use them correctly?

Project Evaluation Rubric

Data Loading and
Model Initialization

Defining Tools

Defining the Agent

Test Cases

Conclusions and
Business
Recommendations

Presentation/Notebook - Overall Quality

What to Cover?

Tool to fetch:

- a patient's profile from the database
- the details of a patient's past encounters from the database
- the clinical notes for a patient from the database
- the details of common trusted medical information sources from the CSV file

Guiding Questions and Considerations

How do we clearly define the inputs and outputs for each tool so the AI agent can call and interpret them reliably?

How do we structure tool outputs in a uniform format so the AI agent can easily combine information from multiple tools?

How do we uniquely identify the patient to avoid any mismatches?

Project Evaluation Rubric

Data Loading and
Model Initialization

Defining Tools

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Presentation/Notebook - Overall Quality

What to Cover?

Tool to fetch:

- a patient's medication details from the database
- a patient's allergy details from the database
- the details of common lab tests from the csv file
- the details of common medications from the CSV file
- a patient's lab test details from the database

Guiding Questions and Considerations

How do we filter encounters (e.g., by time range) based on the agent's task?

How do we structure each source's details?

Project Evaluation Rubric

Data Loading and
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Defining Tools

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Presentation/Notebook - Overall Quality

What to Cover?

Define the Agent State

Define the ReAct prompt

Guiding Questions and Considerations

How do we decide what information the agent needs to keep in its state?

How do we handle initialization, reset, and cleanup of the state?

Is the ReAct prompt structured to enforce Thought -> Action -> Observation -> Final Answer flow?

Project Evaluation Rubric

Data Loading and
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Presentation/Notebook - Overall Quality

What to Cover?

Define the Agent Nodes

Define the Agent Workflow

Visualize the Agent Workflow

Guiding Questions and Considerations

How is state maintained across turns?

What is the starting point of the workflow?

How do we decide when the agent should call a tool, which specific tool to use, and in what order?

Does the diagram accurately reflect the implemented logic without oversimplification?

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Project Evaluation Rubric

Data Loading and
Model Initialization

Defining Tools

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Test Cases

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Presentation/Notebook - Overall Quality

What to Cover?

Execute and print the AI Agent responses for all test cases

Print the trace of AI Agent messages for all test cases

Comment on the accuracy and relevance of the AI Agent response

Guiding Questions and Considerations

Did the agent execute correctly with appropriate tool usage and logical reasoning flow?

Does the trace of AI Agent messages capture the full reasoning flow?

Is the final response accurate, relevant, and fully aligned with the test case requirements?

Project Evaluation Rubric

Data Loading and
Model Initialization

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Presentation /
Notebook Quality

What to Cover?

Conclude with the key takeaways
for the business

Guiding Questions and Considerations

What are the key things that we learned from the data that's useful for
the business?

Logical next steps are a key inclusion

Project Evaluation Rubric

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Presentation /
Notebook Quality

Low-Code Considerations

Clear structure, flow, and visual appeal - everything sits well in a story

Crispness - Focus on key points

All sections of the rubric included, even if in an appendix

Full-Code Considerations

Clear structure and flow

Well-commented and executed code

All sections of the rubric included, even if in an appendix

Project Submission Guidelines: Low-Code

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

Project Submission Guidelines: Low-Code

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

Project Submission Guidelines: Full-Code

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Please feel free to

- leverage the pre-filled code blocks as they are, or
- update the pre-filled code blocks to incorporate necessary changes as per your desired solution workflow for the business problem at hand, or
- discard the pre-filled code blocks and write the entire code from scratch

Project Submission Guidelines: Full-Code

Clearly write down observations, insights, and recommendations for the business problem based on the analysis

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via [free online tools](#)

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission



How do I structure prompts for better medical explanations?

Effective prompt structure typically includes:

- Role definition
- Clear task instruction
- Data context (structured)
- Safety boundaries
- Tone requirement

Unstructured raw clinical text often produces unstable outputs. It's important to pre-format and label fields explicitly.

Why am I getting errors while parsing or loading clinical data?

Common issues include:

- Inconsistent column names
- Mixed data types (e.g., strings in numeric fields)
- Date formatting inconsistencies
- Improper key relationships between tables

It's important to validate schemas early, and standardize formats before using data in downstream logic.



Power Ahead!

